

SUCCESS SECRETS WITH ALEX

MAKE MONEY WITH AI CHALLENGE | தமிழ்

DAY 3 MINI-PROJECT | Your First AI Operating System

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[STEP 1] YOUR PROMPT (From the Questionnaire):

Generate a Markdown file '[GEMINI.md](http://gemini.md)' that I can copy and paste into Antigravity to Build a Dashboard with all the details below.

✓ *What is the Name of the Business?*

Melody Mart

✓ *What is the Business About?*

Melody Mart is an online musical instrument store and service management platform. It allows users to browse, buy, and rent various musical instruments, listen to audio samples before purchasing, and book instrument repair or tuning services online.

✓ *What Products/ Services do the Business Offer?*

-Products:

- String Instruments: Acoustic & Electric Guitars, Ukuleles, Violins, Veena.
- Keyboards & Percussion: Digital Pianos, Synthesizers, Drum Kits, Tabla.
- Accessories: Amplifiers, Cables, Guitar Picks, Tuners, Carrying Cases.

- Services:

- *Instrument Rentals: Short-term rentals for events and beginner practice.*
- *Repair & Restoration: Guitar re-stringing, keyboard key repairs, and electronic servicing.*
- *Tuning & Maintenance: On-demand professional instrument setup.*

✓ What Functionalities do you want the Dashboard to have?

- Admin Dashboard:

- *Inventory Management: Add, update, or remove instruments and track stock levels.*
- *Order Tracking: View pending, completed, and canceled customer orders.*
- *Rental Management: Track rented items, due dates, and return statuses.*
- *Service Bookings: Manage repair requests and technician schedules.*
- *Sales Analytics: Total revenue charts and top-selling instruments.*

- Customer Dashboard:

- *My Orders & Rentals: View current orders, delivery status, and active rental items.*
- *Service History: Track status of booked repair/tuning services.*
- *Wishlist & Cart: Saved items and active cart items.*

✓ What Brand Guidelines do you want the Dashboard to be based off?

- *Primary Color: Deep Midnight Blue (`#1E1E2F`) – for a sleek, premium acoustic feel.*
- *Secondary/Accent Color: Warm Amber/Gold (`#F39C12`) – representing warmth, classic wooden instruments, and creativity.*
- *Background: Light Gray/Off-White (`#F4F6F9`) for high readability.*
- *Typography: Modern Sans-Serif fonts like *Inter* or *Poppins* for clean navigation.*

✓ What Design Preferences do you want the Dashboard to be based on?

- *Style: Modern, minimal, and clean UI with subtle shadows and card layouts.*
- *Layout: Left sidebar navigation for easy module switching, top navbar with search and notifications, and a main content grid area.*
- *Visual Elements: Interactive data tables, visual metric cards (KPIs), quick status badges (e.g., In Stock, Rented, Pending), and audio preview widgets for product pages.*

✓ What Tech Stack to use to build the Dashboard?

You come up with the best

[STEP 2] YOUR [GEMINI.MD](#) File:

1. ROLE & OBJECTIVE

You are the lead product designer, senior frontend engineer, and full-stack engineer responsible for building Melody Mart.

You are an autonomous coding agent.

Your job is to build the actual working application, not just create a visual mockup.

You must:

- Inspect the existing project before changing anything.
- Reuse existing architecture where appropriate.
- Choose sensible technical solutions when requirements are unclear.
- Build reusable, maintainable components.
- Use realistic business data.
- Make all important interactions functional.
- Test and fix issues continuously.
- Do not stop after creating a basic dashboard.
- Continue polishing until the application feels production-ready.

Prioritize:

1. UX

2. Visual quality
3. Functionality
4. Maintainability
5. Performance
6. Accessibility

2. BUSINESS

Melody Mart

Melody Mart is an online musical instrument store combined with rental and instrument-service management.

Customers can:

- Browse instruments and accessories.
- Listen to audio samples.
- Buy instruments.
- Rent instruments.
- Book repair/tuning services.
- Track orders.
- Track rentals.
- Track service bookings.
- Manage wishlist and cart.

Administrators can:

- Manage products and inventory.
- Manage orders.
- Manage rentals.
- Manage repair/service bookings.
- Schedule technicians.
- Manage customers.
- Monitor sales and business analytics.

The product should feel:

Premium + Warm + Musical + Modern + Minimal + Trustworthy

3. INSPECT THE EXISTING PROJECT FIRST

Before coding:

1. Inspect the repository.
2. Read `package.json`.
3. Identify the framework and routing system.
4. Inspect the existing folder structure.
5. Check existing UI components and styling.
6. Check authentication.
7. Check database/backend setup.
8. Check existing dependencies.
9. Determine what is already implemented.

Do not replace a working architecture unnecessarily.

If the project is empty or unsuitable, use the recommended stack below.

4. TECH STACK

Preferred stack:

- Next.js
- TypeScript
- Tailwind CSS
- shadcn/ui
- Lucide React
- Recharts
- React Hook Form
- Zod
- TanStack Table
- Supabase, when backend functionality is required

Use Next.js App Router.

Use Supabase for:

- PostgreSQL
- Authentication

- Storage
- Row Level Security

If backend configuration is unavailable, create a clean mock-data repository/service layer.

The UI must not be tightly coupled to mock data.

5. ENGINEERING RULES

Use:

- Strict TypeScript.
- Reusable components.
- Separation of UI, business logic, validation, and data access.
- Typed models and functions.
- Server Components where appropriate.
- Efficient data fetching.
- Image optimization.
- Pagination where needed.

Avoid:

- `any`
- Giant components.
- Duplicated UI.
- Business logic scattered inside components.
- Unnecessary dependencies.
- Hardcoded values directly inside UI components.

Suggested structure:

```
app/  
components/  
  layout/  
  dashboard/  
  products/  
  orders/  
  rentals/  
  services/  
  customers/
```

```
analytics/  
audio/  
ui/  
lib/  
db/  
repositories/  
services/  
validations/  
utils/  
types/  
data/
```

6. BRAND & DESIGN SYSTEM

Primary

#1E1E2F

Deep Midnight Blue.

Use for:

- Sidebar
- Primary buttons
- Navigation
- Important headings
- Selected states

Accent

#F39C12

Warm Amber/Gold.

Use for:

- CTAs
- Active states
- Audio controls
- Important metrics
- Charts
- Highlights

Background

#F4F6F9

Supporting colors:

White: #FFFFFF
Text: #1E1E2F
Muted: #6B7280
Border: #E5E7EB
Success: #16A34A
Warning: #F59E0B
Danger: #DC2626
Info: #2563EB

Use **Inter** as the main font.

Poppins may be used sparingly for display headings.

7. VISUAL STYLE

Create a modern premium SaaS/e-commerce interface.

Use:

- White cards
- Subtle borders
- Soft shadows
- Rounded corners
- Generous whitespace
- Strong typography hierarchy
- Clean tables
- KPI cards
- Status badges
- Smooth micro-interactions

Avoid:

- Excessive gradients
- Neon colors
- Heavy glassmorphism
- Cartoon styling
- Excessive animations
- Generic Bootstrap/template appearance
- Emojis as UI icons

Use Lucide icons.

The design must feel specifically created for a musical instrument business.

8. APPLICATION EXPERIENCES

Build two primary experiences:

```
/admin  
/account
```

Also create the public shopping/service experience:

```
/shop  
/shop/[productId]  
/services  
/cart  
/checkout  
/login  
/register
```

Authentication and authorization should determine whether the user is:

- Admin
 - Customer
 - Technician
-

9. ADMIN NAVIGATION

Create a responsive sidebar:

Melody Mart

Dashboard

Management

Inventory

Products

Orders

Rentals

Service Bookings

Customers

Insights

Analytics

System

Notifications

Settings

Sidebar must support:

- Active states
- Hover states
- Collapsed desktop mode
- Mobile drawer
- User profile

Top navbar should contain:

- Global search
 - Notifications
 - User profile
 - Page context
-

10. ADMIN DASHBOARD

Route:

/admin

Create a polished business overview.

Header:

Good evening, Admin
Here's what's happening with Melody Mart today.

Include date filters:

- Today
- Last 7 Days
- Last 30 Days
- Last 3 Months
- Custom

KPI cards:

- Total Revenue
- Orders
- Active Rentals
- Service Bookings
- Low Stock
- Pending Orders
- Use realistic INR data.

Values should come from the data layer.

11. ADMIN ANALYTICS

Create:

Revenue Overview

Show:

- Product sales
- Rental revenue
- Service revenue

Use Recharts.

Sales by Category

Categories:

- Guitars
- Keyboards
- Percussion
- String Instruments
- Accessories

Top Selling Products

Show:

- Product
- Category
- Units sold
- Revenue
- Stock
- Trend

Recent Orders

Show:

- Order ID
- Customer

- Product
- Date
- Amount
- Payment
- Status
- Actions

Use semantic status badges.

12. INVENTORY & PRODUCTS

Routes:

/admin/inventory
/admin/products

Inventory must support:

- Search
- Filter
- Sort
- Add
- Edit
- Delete
- Update stock
- View product
- Rental status

Product fields:

Name
SKU
Brand
Category
Description
Selling Price
Rental Price
Discount
Tax
Stock
Reorder Threshold
Location
Images
Thumbnail
Audio Sample

Instrument Type
Material
Skill Level
Color
Weight
Dimensions
Available for Sale
Available for Rent
Available for Service

Stock statuses:

In Stock
Low Stock
Out of Stock
Reserved
Rented
Under Repair

Forms must use React Hook Form + Zod validation.

13. AUDIO PREVIEW

Audio preview is a **core Melody Mart feature**.

Create a reusable:

AudioPlayer

It must support:

- Play/pause
- Progress
- Current time
- Duration
- Volume
- Instrument name
- Waveform-style progress

Example:

► —●— 0:24 / 1:12

Yamaha Acoustic Guitar
Warm Tone Demo

Use amber for playback.

Only one audio preview should play at a time.

If real audio files are unavailable, use safe demo audio without breaking the interface.

14. ORDER MANAGEMENT

Route:

/admin/orders

Support:

- Search
- Filters
- Sorting
- Pagination
- View order
- Update status
- Cancel eligible orders

Order detail should contain:

Customer

- Name
- Email
- Phone
- Address

Items

- Product
- Quantity
- Unit price
- Total

Payment

- Method
- Status
- Transaction reference

Delivery

- Shipping status
- Tracking
- Estimated delivery

Create an order timeline:

Order Placed

→ Confirmed

→ Packed

→ Shipped

→ Delivered

Statuses:

Pending

Processing

Shipped

Delivered

Cancelled

15. RENTAL MANAGEMENT

Route:

/admin/rentals

Track:

- Rental ID
- Customer
- Instrument
- Start date
- Due date
- Return date
- Rental fee
- Deposit
- Condition
- Status

Statuses:

Reserved

Active

Due Soon

Overdue
Returned
Damaged

Show summary cards:

- Active
- Due Today
- Due This Week
- Overdue
- Returned This Month

Business logic:

today > due date + not returned = Overdue

due within 2 days = Due Soon

Overdue rentals should be visually prominent.

16. SERVICE MANAGEMENT

Route:

/admin/services

Services:

- Guitar restringing
- Keyboard repair
- Electronics servicing
- Instrument tuning
- Instrument setup
- Restoration

Booking information:

- Booking ID
- Customer
- Instrument
- Service
- Date
- Time
- Technician
- Estimated cost
- Status
- Actions

Statuses:

Requested
Confirmed
Assigned
In Progress
Ready for Pickup
Completed
Cancelled

17. TECHNICIAN MANAGEMENT

Create technician scheduling.

Technician fields:

- Name
- Specialty
- Email
- Phone
- Availability
- Current jobs
- Rating

Specialties:

- Guitar Technician
- Keyboard Specialist
- Electronics Technician
- String Instrument Specialist

Allow:

- Assign technician
- Reschedule
- View daily schedule
- View weekly schedule

Use a clean calendar/scheduling UI.

18. CUSTOMER MANAGEMENT

Route:

/admin/customers

Display:

- Name
- Email
- Phone
- Orders
- Rentals
- Services
- Total spent
- Joined date
- Status

Customer detail should include:

- Profile
- Orders
- Rentals
- Service history
- Wishlist
- Spending

19. CUSTOMER DASHBOARD

Route:

/account

Customer navigation:

Overview
My Orders
My Rentals
Service Bookings
Wishlist
Cart
Notifications
Profile

Dashboard header:

Welcome back, Alex
Your instruments and services at a glance.

Show:

- Active Orders
- Active Rentals

- Upcoming Services
- Wishlist Items

Also show:

- Recent orders
- Active rentals
- Upcoming services

Keep the customer interface simpler than the admin interface.

20. SHOP & PRODUCT EXPERIENCE

Routes:

/shop
/shop/[productId]

Categories:

String Instruments

- Acoustic Guitars
- Electric Guitars
- Ukuleles
- Violins
- Veena

Keyboards & Percussion

- Digital Pianos
- Synthesizers
- Drum Kits
- Tabla

Accessories

- Amplifiers
- Cables
- Guitar Picks
- Tuners
- Carrying Cases

Product detail must include:

- Image gallery
- Product name
- Brand
- Rating
- Price
- Rental price
- Stock
- Quantity
- Add to Cart
- Rent
- Wishlist
- Description
- Specifications
- What's Included
- Shipping
- Rental information
- Warranty
- Reviews
- Audio Preview

Important section:

Listen Before You Buy

Use the reusable `AudioPlayer`.

21. CART & WISHLIST

Cart route:

/cart

Support:

- Product image
- Name
- Price
- Quantity
- Subtotal
- Remove
- Save for later

Summary:

Subtotal
Discount
Shipping
Tax
Total

CTA:

Proceed to Checkout

Use INR formatting.

Wishlist route:

/account/wishlist

Product cards should include:

- Image
- Name
- Brand
- Price
- Rating
- Stock
- Audio preview
- Add to cart
- Remove wishlist

22. CUSTOMER ORDERS, RENTALS & SERVICES

Customer routes:

/account/orders
/account/rentals
/account/services

Orders

Allow:

- View order
- Track delivery
- View invoice
- Reorder
- Cancel eligible orders

Rentals

Show:

- Instrument
- Rental period
- Start date
- Due date
- Days remaining
- Deposit
- Status
- Return instructions

Services

Show:

- Instrument
- Service
- Date
- Technician
- Status
- Cost
- Notes

Use timelines for orders, rentals, and service progress where appropriate.

23. SERVICE BOOKING FLOW

Route:

/services

Build a multi-step booking flow:

Step 1

Select instrument.

Step 2

Select service.

Step 3

Select date/time.

Step 4

Enter issue/details.

Allow image uploads if backend supports them.

Step 5

Confirmation.

Show:

- Booking ID
- Instrument
- Service
- Date
- Time
- Technician
- Estimated cost

24. DATABASE & DATA ARCHITECTURE

If Supabase is available, create tables for:

users
products
orders
order_items
rentals
service_bookings
technicians
wishlist_items
cart_items

Important product fields:

id
name
sku
brand
category
description
price
rental_price
stock_quantity

```
reorder_threshold
condition
image_url
audio_url
available_for_sale
available_for_rent
created_at
updated_at
```

Use repositories/services such as:

```
getProducts()
getOrders()
getRentals()
getServiceBookings()
getCustomers()
getAnalytics()
```

Never scatter database queries directly throughout UI components.

If no backend exists, create realistic mock data:

- 20+ products
- 10+ customers
- 15+ orders
- 8+ rentals
- 10+ service bookings
- 5+ technicians
- 10+ wishlist items

Use realistic Indian names and INR pricing.

Never use:

```
Product 1
User 123
Lorem ipsum
Test Product
```

25. GLOBAL SEARCH & NOTIFICATIONS

Implement global search across:

- Products
- Orders
- Customers
- Rentals
- Services

Search should support:

- Suggestions
- Recent searches
- Categories
- Keyboard navigation
- Empty states

Notifications should support:

- Order updates
- Rental reminders
- Service updates
- Low-stock alerts
- Promotions

Unread notifications must be visually distinct.

26. UX, STATES & INTERACTIONS

Every important action must actually work.

Implement:

- Navigation
- Search
- Filters
- Sorting
- Pagination
- Add/edit/delete
- Stock updates
- Order status updates
- Rental status updates
- Technician assignment
- Service booking
- Wishlist toggle
- Cart quantity changes
- Remove cart items
- Audio playback
- Date filters
- Dialogs
- Dropdowns
- Notifications
- Logout

Use toast feedback for successful actions.

Examples:

Product added successfully.

Product updated.

Order status updated.

Rental marked as returned.

Technician assigned.

Service booking confirmed.

Added to wishlist.

Added to cart.

Create proper:

- Loading states
- Skeletons
- Empty states
- Error states
- Form validation
- Retry actions

Never leave a blank screen or broken button.

27. RESPONSIVE & ACCESSIBLE DESIGN

Design intentionally for:

320px

375px

768px

1024px

1280px+

Desktop:

Sidebar + Topbar + Content

Tablet:

- Collapsible sidebar
- Responsive grids

Mobile:

- Top navigation
- Drawer navigation
- Single-column layouts
- Responsive cards
- Scrollable tables when necessary

Check for:

- Horizontal overflow
- Text clipping
- Modal overflow
- Chart overflow
- Broken tables
- Image distortion

Accessibility requirements:

- Semantic HTML
- Keyboard navigation
- Visible focus states
- Accessible labels
- ARIA where needed
- Good color contrast
- Status should not rely only on color
- Accessible dialogs/dropdowns/tables
- Respect `prefers-reduced-motion`

28. SECURITY & PERFORMANCE

If using Supabase:

- Enable Row Level Security.
- Customers can only access their own data.
- Technicians only access relevant service information.
- Admins can access business data.
- Protect admin routes.
- Validate inputs.
- Keep secrets server-side.
- Use environment variables correctly.

For performance:

- Use Server Components where appropriate.
- Optimize images.
- Lazy-load expensive components.
- Paginate large datasets.
- Avoid unnecessary client-side JavaScript.
- Avoid unnecessary dependencies.

29. IMPLEMENTATION STRATEGY

Build in this order:

Phase 1 — Foundation

- Project setup
- Theme
- Typography
- Global styles
- Layout
- Sidebar
- Navbar
- Responsive shell
- Reusable UI components

Phase 2 — Admin

1. Dashboard
2. Inventory
3. Products
4. Orders
5. Rentals
6. Services
7. Customers
8. Analytics

Phase 3 — Customer

1. Dashboard
2. Shop
3. Product detail
4. Orders
5. Rentals
6. Services
7. Wishlist
8. Cart
9. Checkout

Phase 4 — Functionality

- Forms
- Validation
- Search
- Filters
- Sorting
- Status updates

- Audio player
- Notifications
- Authentication
- Role-based access

Phase 5 – Backend

If required:

- Supabase
- Database
- Authentication
- Storage
- RLS
- Repositories

Phase 6 – Polish

- Responsive QA
- Accessibility
- Performance
- Visual refinement
- Error handling
- Loading states
- Empty states
- Final cleanup

30. QUALITY CONTROL & DEFINITION OF DONE

After every major feature:

1. Run the application.
2. Inspect it visually.
3. Test interactions.
4. Check console errors.
5. Check TypeScript errors.
6. Check build errors.
7. Test responsive behavior.
8. Fix problems immediately.

Before considering the project complete:

Admin

- Dashboard works
- Inventory works
- Products work
- Orders work
- Rentals work
- Services work
- Customers work
- Analytics work

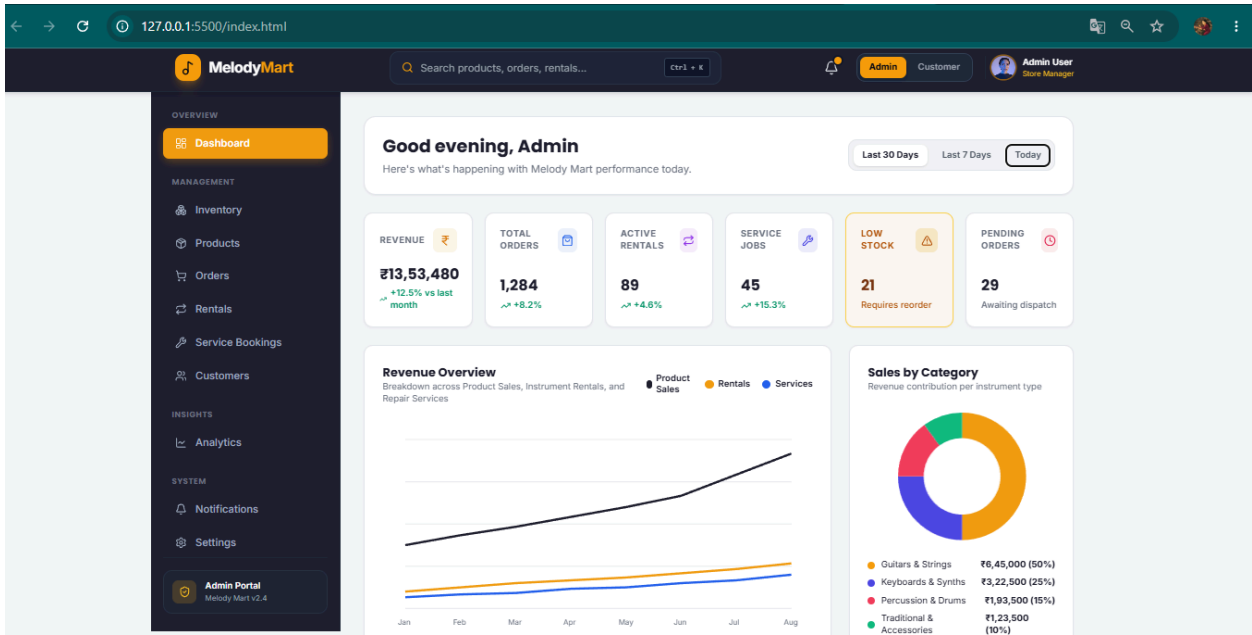
Customer

- Shop works
- Product details work
- Audio preview works
- Cart works
- Wishlist works
- Orders work
- Rentals work
- Services work

Final checks

- No broken routes.
- No dead buttons.
- No obvious console errors.
- No TypeScript errors.
- Forms validate.
- Search works.
- Filters work.
- Status updates work.
- Mobile works.
- Accessibility has been considered.
- UI feels premium and consistent.

[STEP 3] YOUR AIOS SCREENSHOTS:



The screenshot displays the MelodyMart Inventory Management page. The left sidebar is identical to the dashboard. The main content area is titled 'Inventory Management' and includes a '+ Add New Stock' button. A search bar allows filtering by product or SKU, and a dropdown menu shows 'All Categories'. The total items are 10, with 3 items in low stock. The table below lists the inventory items with their details and actions.

INSTRUMENT	SKU	CATEGORY	PRICE	STOCK LEVEL	STATUS	CONDITION	ACTIONS
 Yamaha F310 Acoustic Guitar	GTR-YAM-F310	String Instruments	₹11,490	14	In Stock	New	 
 Fender Player Stratocaster	GTR-FND-STRAT	String Instruments	₹68,500	4	Low Stock	New	 
 Yamaha P-125 Digital Piano	KEY-YAM-P125	Keyboards & Synthesizers	₹54,990	7	In Stock	New	 
 Roland JUNO-DS61 Synthesizer	KEY-ROL-JUNO	Keyboards & Synthesizers	₹64,900	2	Low Stock	New	 
 Pearl Export EXX 5-Piece Drum Kit	PER-PRL-	Percussion	₹78,900	3	In Stock	New	 

